7 **(a)** The emission spectrum of atomic hydrogen consists of a number of discrete wavelengths. Explain how this observation leads to an understanding that there are discrete electron energy levels in atoms.

....

....

....

.... [2]

Discrete wavelength is associated with a discrete energy of the photon.

Discrete energy of the photon imply discrete energy <u>changes</u> / difference which implies discrete energy levels

The key point is that discrete wavelengths imply discrete energies of the photons (not energy levels of the photons). This was not always stated explicitly. Formulas were used without further explanation, requiring the examiner to make inferences.

Then because the energy of the photon is discrete, this must imply discrete energy changes for the electron hence implying discrete energy levels.

Many responses made correct statements but lacked the requisite flow of logic, e.g., discrete wavelengths imply discrete energy levels. The focus of the question is to make logical inferences and not merely describe what is known about quantum energy levels.

(b) Three electron energy levels in atomic hydrogen are represented in Fig. 7.1.

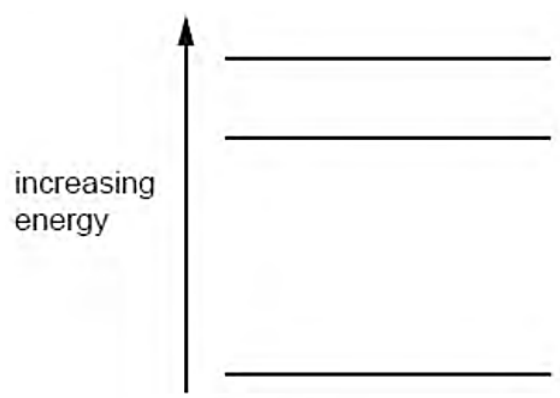

Fig. 7.1

The wavelengths of the spectral lines produced by electron transitions between these three energy levels are 486 nm, 656 nm and 1880 nm.

(i) On Fig. 7.1, draw arrows to show the electron transitions between the energy levels that would give rise to these wavelengths.
Label each arrow with the wavelength of the emitted photon. [3]

Three energy changes shown correctly
Arrows ‘pointing’ in correct direction (decreasing energy)
Wavelengths correctly identified

This part was well done.

(ii) Calculate the maximum change in energy of an electron when making transitions between these levels.

[Turn over

			energy = ………………………………… J	[2]
			$\Delta E = hc / \lambda$ $= (6.63 \times 10^{-34})(3.00 \times 10^{8}) / (486 \times 10^{-9}) = 4.09 \times 10^{-19}$ J *This part was well done.*	
		(iii)	When an electron undergoes transition, there is a change in momentum of the hydrogen atom.	
			1. Explain the origin of the change in momentum of the hydrogen atom. ………………………………………………………………………………………… ………………………………………………………………………………………… …………………………………………………………………………………………	[2]
			Since photon has momentum, photon emission comes with a change in momentum Since total momentum conserved, equal and opposite change in momentum of hydrogen atom *The key point is to discuss the energy or momentum of the photon. Many responses simply state that the atom must lose or gain energy without further explanation. The source or origin of this change in energy should be explained.* *Also, students should not assume that the initial total momentum is zero. Unnecessary or erroneous assumptions can be penalized.*	
			2. For the electron transition in **(ii)**, calculate the change in momentum of the hydrogen atom. change in momentum = ………………………………… kg m s^{-1}	[1]
			(magnitude of) change in momentum = momentum of photon = h / λ $= (6.63 \times 10^{-34}) / (486 \times 10^{-9})$ $= 1.36 \times 10^{-27}$ kg m s^{-1} *This was generally well done, although some attempted to use $p^2/2m$ which leads nowhere.*	
				[Total: 10]

Section B

Answer **one** question from this Section in the spaces provided.

8	In a power station generator, a large rectangular coil is rotating at 50 revolutions per second in a